SPINAL CORD HISTOLOGY

Learning Objectives:

At the end of the practical, the student should be able to:

- Describe white matter.
- Describe Gray Matter.
- Identify structures in the gray and white matter.

White Matter :

- Anterior Funiculus (Anterior White Column)
- Posterior Funiculus (Posterior White Column)
- *Fasciculus Gracilis & Fasciculus Cuneatus*
- Lateral Funiculus (Lateral White Column).

Gray Matter:

- Anterior Horn ---------- --- motor.
- Posterior Horn ---------- sensory.
- Lateral Horn ---------- autonomic (sympathetic).
- Gray Commissure -------- anterior and posterior.

1. Posterior horn.

2. Anterior horn.

3. Intermediate zone (intermediate gray).

4. Lateral horn.

5. Posterior funiculus.

6. Anterior funiculus.

7. Lateral funiculus.

8. Lissauer's tract.

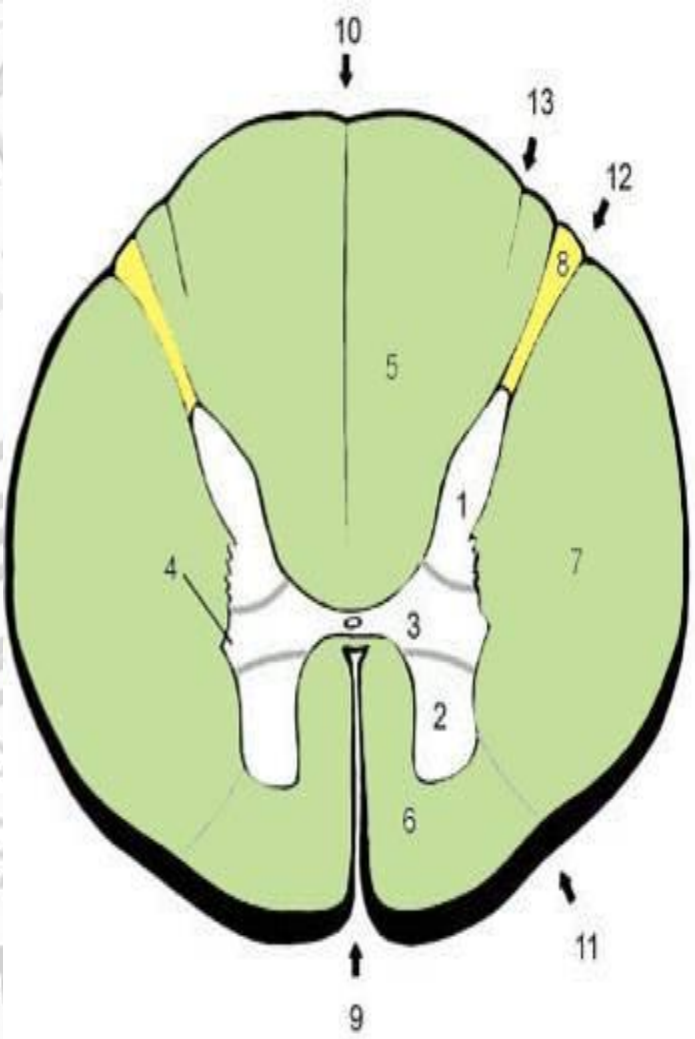
9. Anterior median fissure.

10. Posterior median sulcus.

11. Anterolateral sulcus.

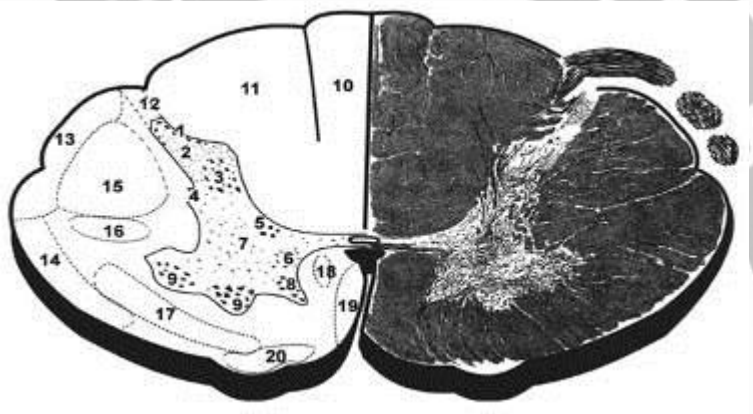
12. Posterolateral sulcus.

13. Posterior Intermediate sulcus.

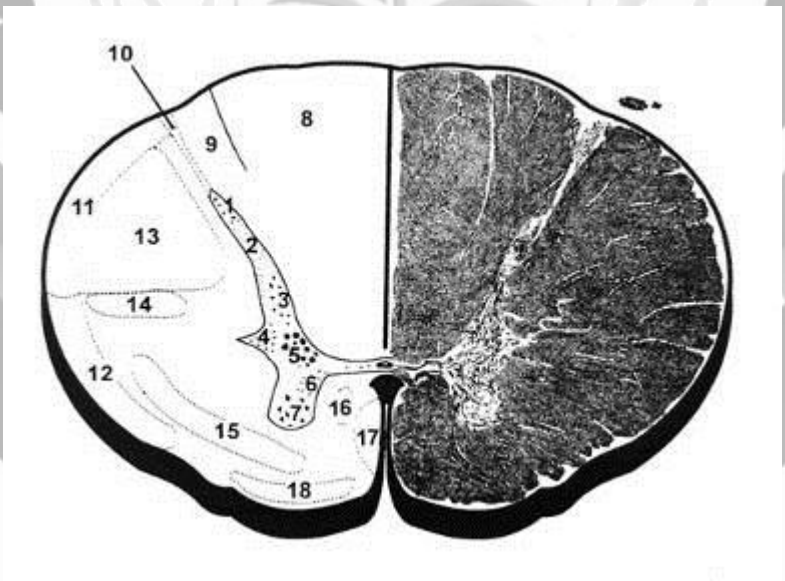


<u>GRAY AND WHITE MATTER VARIES AT DIFFERENT SEGMENTAL LEVELS:</u>

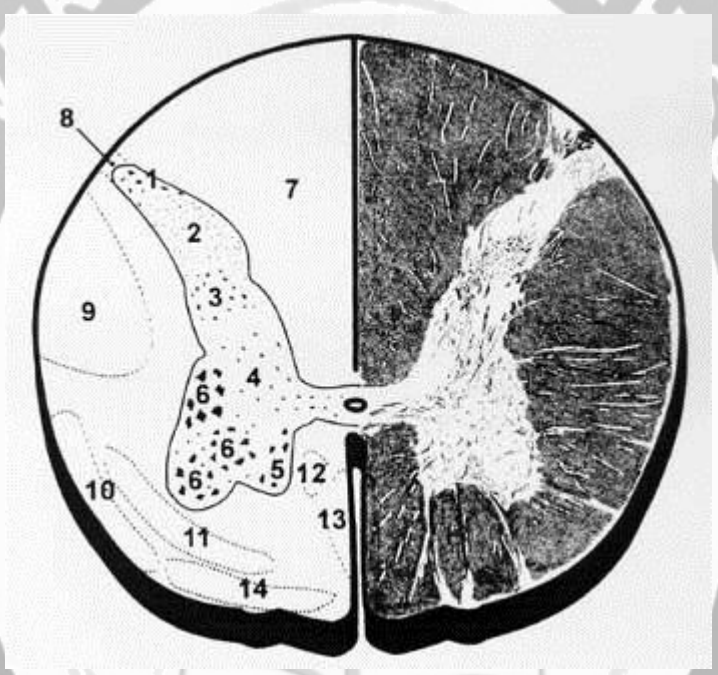
Cervical Enlargement (C8).



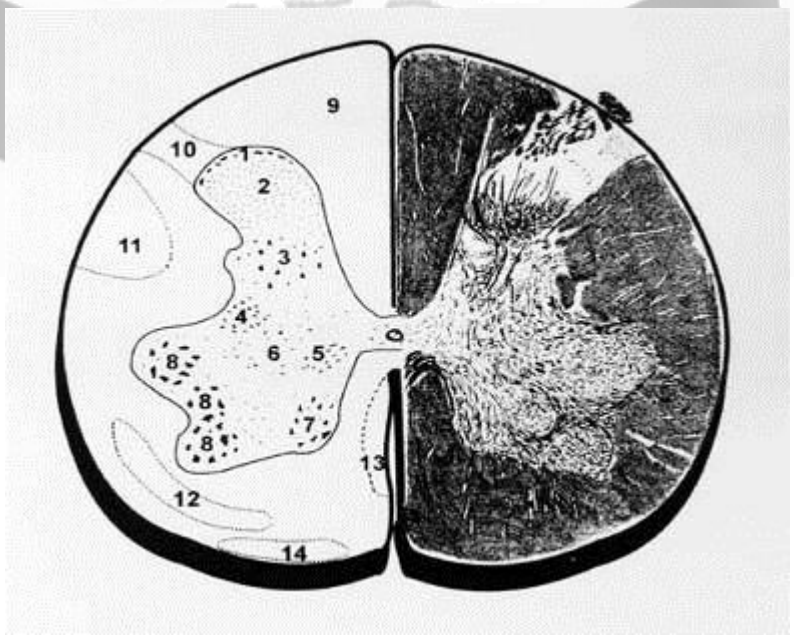
Thoracic Cord (T8)



Lumbar Enlargement (L3)



Sacral Cord (S1)



Cervical level :

Wide flat cord, lots of white matter, ventral horn enlargements**.**

Lumbar level:

- Round cord, ventral horn enlargements.

Sacral level:

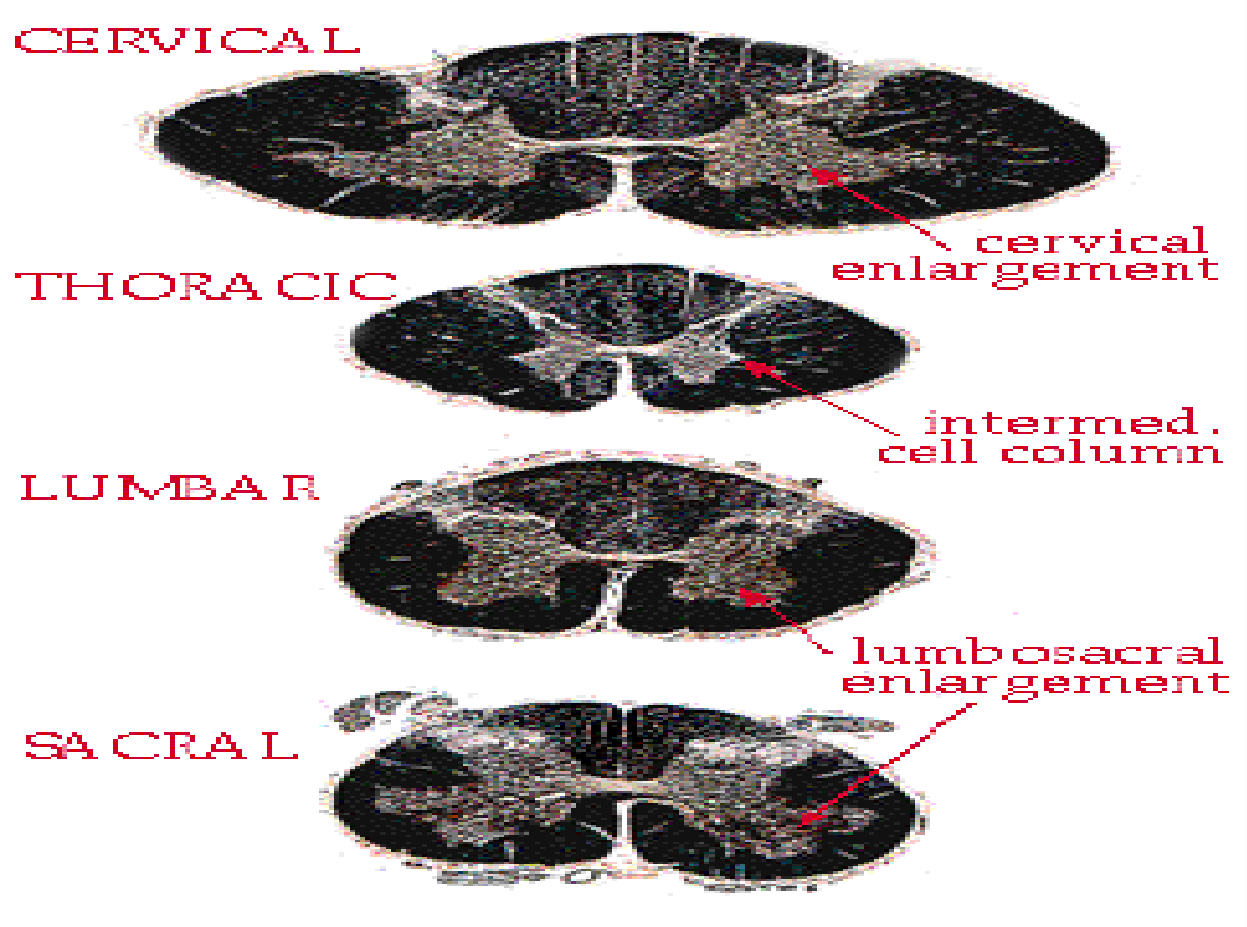
- Small round cord, lateral Horn.

Thoracic level:

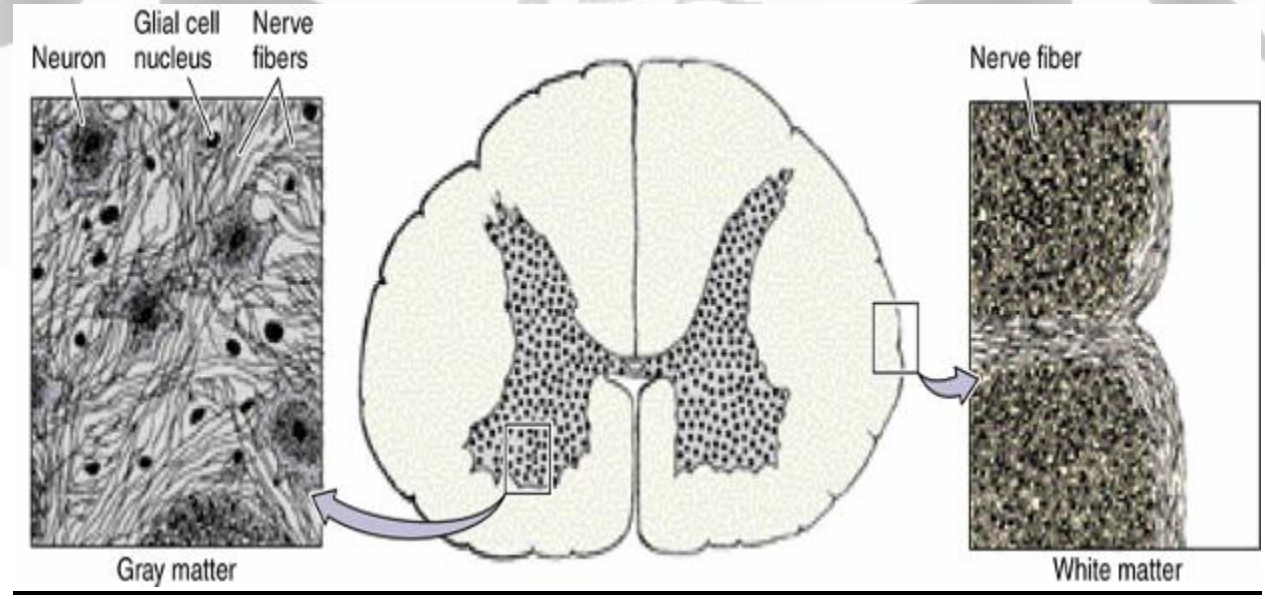
- Notice the pointed tips which stick out between the small dorsal and ventral horns.

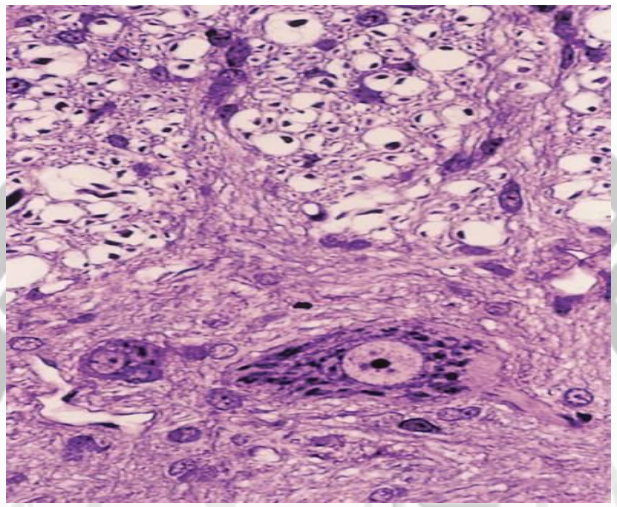
This extra cell column is called the intermediate horn *Intermediolateral Cell Column*).

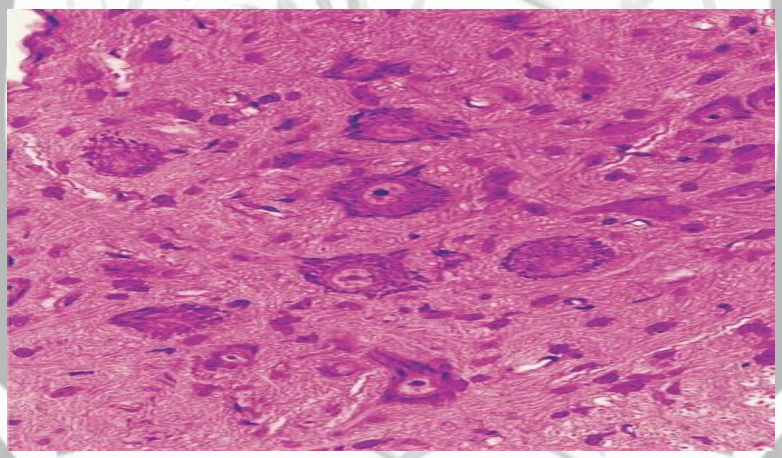
It is the source of all of the sympathetic in the body & occurs only in the Thoracic sections T 1 to L 2-L3.



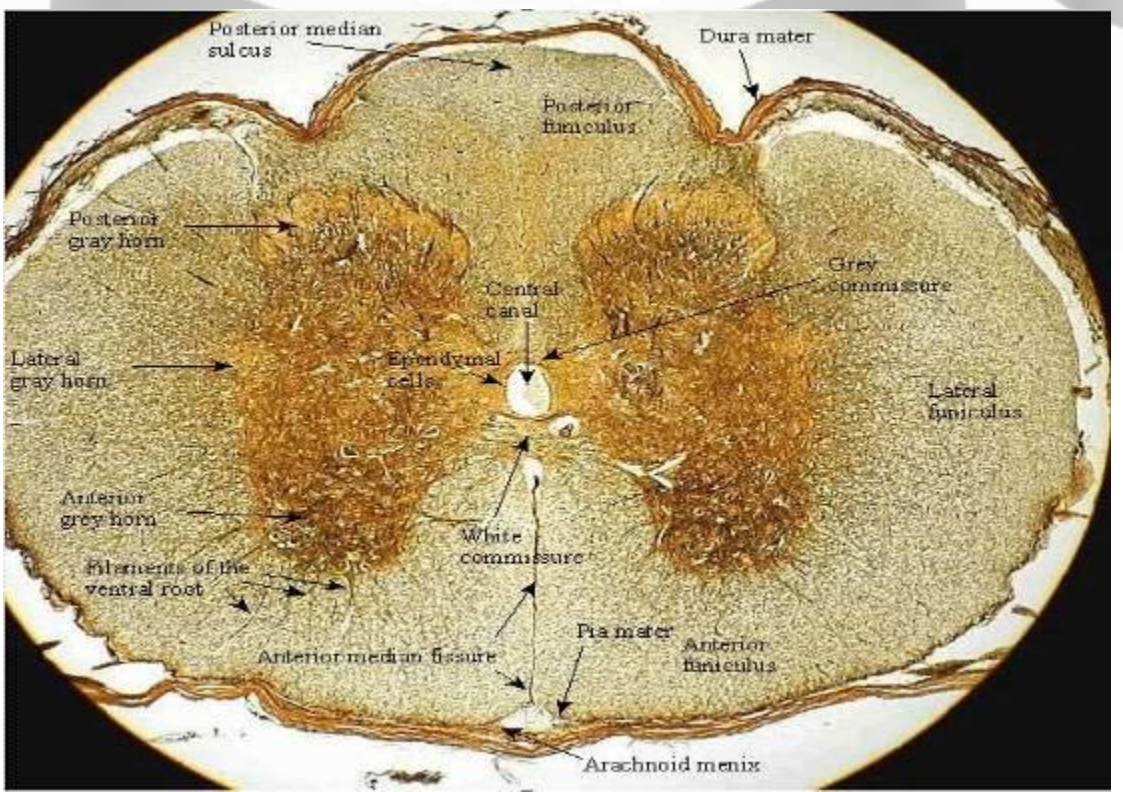
COMPOSIT STRUCTURES IN GRAY AND WHITE MATTER:





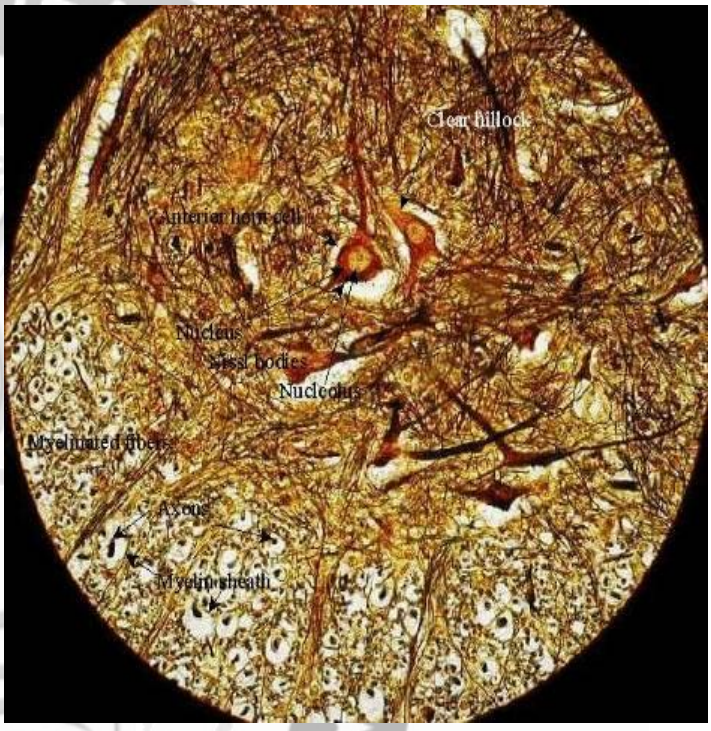


Overview – Cross section of Spinal Cord



Anterior horn cell (400x)

- clear hillock
- nucleus
- nucleolus
- Nissl bodies



MYELINATED FIBERS

(cross section at posterior funiculus)
multiple cross sections of fibers
myelin sheath axon.

